MIL-D-2351C

9 July 1968
SUPERSEDING
MIL-D-2351B
31 March 1966

MILITARY SPECIFICATION

DOUGH INTERMEDIATE PROOFING MACHINES

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers electrically operated dough intermediate proofing machines.

* 1.2 Classification.- The machines covered by this specification shall be the following types, styles and sizes, as specified (see 6.2).

Type I - L-shaped
Type II - T-shaped

STYLES

	FEED (to side or front of loader)	DISCHARGE
Style 1	Right hand	Right hand
Style 2	Left hand	Right hand
Style 3	Front	Right hand
Style 4	Right hand	Left hand
Style 5	Left hand	Left hand
Style 6	Front	Left hand

SIZES
(No. of active dough-holding pockets)

Size 1	350 to 375
Size 2	450 to 475
Size 3	530 to 565
Size 4	590 to 655

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 7320

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SPECIFICATIONS

FEDERAL

CC-M-636 - Motors, Alternating-Current, (Fractional Horsepower).
CC-M-641 - Motor, Alternating Current (Integral Horsepower 200 HP and smaller).

MILITARY

MIL-B-2316 -- Bakery Equipment, Including Units Assemblies, Repair Parts and Tools, Preparation for Delivery Of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics - Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

BAKING INDUSTRY SANITATION STANDARDS COMMITTEE (BISSC)

SANITATION STANDARD NO. 3: MECHANICAL PROOFERS

(Application for copies should be addressed to the Baking Industry Sanitation Standards Committee, 521 Fifth Avenue, New York, N.Y., 10017.)

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UNDERWRITERS' LABORATORIES, INC. (UL)

Electrical Appliance and Utilization Equipment List

Standard UL 73-1967: Motor Operated Applicances

(Requests for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 1285 Walt Whitman Road, Melville, L.I., New York 11746, or 1655 Scott Boulevard, Santa Clara, Calif. 95050.)

3. REQUIREMENTS

* 3.1 First article.- Unless otherwise specified (see 6.2), before production is commenced, a sample dough intermediate proofing machine shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product.- Dough intermediate proofing machines delivered under this specification shall be the manufacturer's current commercial product, except for any changes necessary to comply with specification requirements.

3.3 Standards.-

3.3.1 BISSC Standard.- Intermediate proofing machines shall conform to the applicable requirements of BISSC Standard No. 3.

3.3.1.1 BISSC Certification.- The supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that intermediate proofing machine he proposes to furnish under this specification meets the applicable requirements of BISSC as follows:

(1) Display of the BISSC seal on the finished intermediate proofing machine shall be submitted prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment or

(2) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the cabinets are being procured, indicating that the cabinets

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have been tested and conform to BISSC Standard No. 3, shall be submitted prior to the approval of the preproduction sample, if one is submitted, or prior to commencing production.

3.3.2 UL Standards.- Electrical components shall conform to the applicable requirements of Standard UL 73, or to the UL Standard for each component.

3.3.2.1 UL Certification.- Prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the dough intermediate proofing machine he proposes to furnish under the specification meet the applicable requirements of Underwriters' Laboratories.

3.3.2.2 UL.- Acceptable evidence of meeting the requirements of the UL shall be the UL Re-examination Service Marker, listing in the UL "Electrical Appliance and Utilization Equipment List" or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the electrical components offered have been tested and conform to the applicable requirements of Standard UL 73, or to the UL Standard for each component.

3.4 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial dough intermediate proofing machine.

3.5 Design.- The dough intermediate proofing machine shall be either type I, "L" shaped or type II, "T" shaped (see 1.2) with the horizontal section above the vertical section for both types. The machine shall be in one of six styles as determined by right, left or front loading and right or left discharge (see 1.2) to permit wide flexibility in bakery lay-out. When tested as specified in 4.4.1 all machine components including dough conveyors, trays, chutes and molder feed rolls, shall function smoothly without binding and without excessive noise. Bearings and other moving parts shall not overheat. The machine shall accommodate dough pieces ranging in weight from 8 to 36 ounces, and the time of the proofing period shall be adjustable from 8 to 16 minutes. The machine shall have ample capacity to proof a full load of 23 ounce dough pieces for one complete cycle, when tested as specified in 4.4.2. Ceiling ties for supporting the horizontal section shall be provided, if necessary.

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3.6 Construction.- The machine shall be constructed of sheet steel over structural steel framework. Hinged or removable panels shall be provided.

* 3.6.1 Loader.- The integral loader shall be so designed that it can be fed from either the right side, left side or front (see 1.2) and shall be equipped with a variable speed control. The loader shall be so designed to receive individual dough pieces from the dough-rounder without doubling, and deposit the pieces automatically into the sub-divided trays of the conveyor in the vertical section of the machine (see 4.4.2). The loader mechanism shall be inclosed and shall be protected by a shear-pin or slip-clutch device.

* 3.6.2 Vertical and horizontal sections.- Unless otherwise specified (see 6.2), the vertical leg section shall be at least 7 feet high. The horizontal section shall be arranged to turn over the dough pieces at least twice (see 4.4.2) before discharge at the opposite side of the vertical leg from the loader.

* 3.6.3 Conveyor.- The conveyor shall include steel roller-chains running with mated sprockets, machined or precision-cast. The conveyor shall have provisions for carrying uniformly-spaced dough trays.

* 3.6.4 Trays.- The trays shall be of metal divided into sections to keep the dough pieces separated in individual pockets. Trays shall be provided with a removable nylon or rust-proof metal liner.

* 3.6.5 Flour duster.- An adjustable flour-dusting device shall be provided on the integral loader and another in the vertical or horizontal section of the machine.

* 3.6.6 Discharge.- An automatic delivery mechanism shall be provided at the dough-discharge point. A slotted or grid swivel chute shall be provided to deposit the uniformly-spaced dough pieces onto the conveyor under the horizontal section of the proofer, for right-hand or left-hand discharge into the molder. A dough-flattening roller shall be provided at the discharge conveyor. Conveyor, chute and flattening roller shall operate efficiently when tested as specified in 4.4.1 and 4.4.2.

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* 3.6.7 Electrical.-

3.6.7.1 Motor.- The drive motor shall conform to style B1 or B2 (totally enclosed) of CC-M-636 or CC-M-641. Unless otherwise specified (see 6.2), the motor shall operate on a nominal 208-volt, 60-hertz, 3-phase alternating current (a.c.) system.

3.6.7.2 Switch.- The motor switch shall be push-button type with thermal overload protection.

3.6.8 Lubrication.- Pressure gun lubrication fittings of the hydraulic type shall be provided at accessible points for all bearings requiring grease lubrication. Closed oil cups shall be provided on all bearings requiring oil lubrication.

3.6.8.1 Lubrication chart.- The supplier shall affix to each machine, a corrosion-resistant metal lubricating chart indicating:

(a) All points of the equipment requiring lubrication.
(b) Grade of lubricant required.
(c) Time interval when lubrication should be accomplished.

3.7 Finish.- All exterior surfaces of the machine, except corrosion-resisting steel, shall be given a white enamel finish, in accordance with the manufacturers commercial practice.

3.8 Marking for identification.- Identification marking shall be in accordance with MIL-STD-130, and on a corrosion-resisting steel plate securely affixed in a conspicuous location on a part of the machine not normally replaced.

3.9 Equipment manual.- Each machine shall be furnished with an equipment manual. Manual contents and additional manual distribution shall be as specified in the contract or order (see 6.2).

* 3.10 Electromagnetic interference control.- Equipment procured for Army shall be designed and equipped for electromagnetic compatibility in accordance with class 11B of MIL-STD-461 (see 4.4.3). When specified (see 6.2), equipment procured for Navy and Air Force shall be in accordance with class 11B of MIL-STD-461 (see 4.4.3).

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3.11 Workmanship.- The finished machine shall be free from defects: marred surfaces, dents, punctures, fractures, rough edges, burrs, slivers. All welds shall be smooth and uniform, completely fused, without undercuts, cracks, burnholes or fractures. Fastener threads shall not be broken or stripped. All fasteners shall be tight. No component shall be missing, malformed. All components shall be carefully assembled, adjusted and aligned.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a preproduction sample is required it shall be examined for the defects in table I and dimensions specified, and tested in accordance with 4.4.1, 4.4.2 unless waived (see 6.3), and 4.4.3, if required (see 6.2).

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 End item inspection.- The inspection lot shall be the number of dough intermediate proofing machines of one type, style and size offered for acceptance at one time. The sample unit for this inspection shall be a completely fabricated dough intermediate proofing machine.

* 4.3.2.1 Visual examination.- Examination of the end item shall be made for defects in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not finished as specified		X
Construction and workmanship general (applicable to all components and assemblies)	Part missing	X	
	Fractured, split, punctured, dented or deteriorated	X	
	Part misplaced, loose, or not in proper alignment	X	
	Sharp burr, sliver, or splinter	X	
	Hinged or removable panels not provided	X	
	Any adjustable assembly that cannot be adjusted	X	
	Any component not accessible for servicing	X	
Loaders	Not equipped with a variable speed control	X	
	Mechanism not inclosed	X	
	Not provided with shear pin or slip clutch device	X	
Vertical leg section	Height not as specified	X	
Horizontal section	No provisions made for turning the dough pieces over	X	
Conve...	Roller chains not provided	X	
	Sprockets with machine-cut or precision-cast teeth not provided	X	
	No provisions for easy removal or replacement of trays	X	
	Not made of metal	X	
	Not provided with rust-proof metal or removable nylon liners		X
	Sections or pockets not provided		X
Flour dusting	Not provided as specified	X	

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Construction and workmanship general (applicable to all components and assemblies) (cont'd)			
Discharge	Delivery not automatic at discharge point of the trays	X	
	Not provided with a slotted or grid swivel chute	X	
	Flattening roller not provided	X	
Welding and brazing (where required)	Missing, incomplete fusion, burr, holes, cracks, fractures, or not otherwise fused	X	
	Slag inclusion, undercut, not smooth and uniform, scale or flux deposits not removed	X	
Soldering (when applicable)	Missing, not adherent or incomplete	X	
	Not clean (surface not finished neatly) or pin holes in soldering		X
	Not clean (flux or flux residue not removed, or pin holes in solder)	X	
Threads	Broken or stripped	X	
Threaded fasteners	Missing, loose or broken	X	
Wiring	Cut, abraded, loose or excessive insulation stripped	X	
Lubrication (when applicable)	Not lubricated where required	X	
	Fittings not accessible for lubrication	X	
	Instructions missing	X	

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Equipment manual	Missing, incomplete. not legible	X	
Marking identification	Missing, incomplete, not legible, not in proper location		X

4.3.2.2 Dimensional examination.- Dimensional examination shall be made of the end item to determine compliance with this specification. Any dimension not within the specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

4.3.2.3 Testing of the end item.- Each production item shall be tested in accordance with 4.4.1. When no preproduction sample is required, the initial item from each lot shall also be tested as specified in 4.4.2 unless waived (see 6.3), and 4.4.3, if required (see 6.2).

4.3.3 Examination of preparation for delivery.- Examination of preservation, packaging, packing, and marking shall be in accordance with MIL-B-2316.

4.3.4 Standards compliance.- Proof of compliance with UL and BISSC requirements of 3.3 shall be available to the Government representative.

4.4 Tests.-

4.4.1 Operation test.- Connect the machine to power and operate for a minimum of one hour without load. While operating, stop and start the machine at least 5 times, and observe bearings for overheating; starting, stopping and conveyor devices, for smooth operation and all moving components for binding or other malfunction, to determine compliance with requirements specified in 3.5.

* 4.4.2 Functional test.- The machine shall be connected to an appropriate power source and operated with a full load of 23 ounce dough pieces for one complete cycle of 8 to 16 minutes, as required. The loader shall be checked for automatic deposition of dough rounds onto the sub-divided trays of the vertical conveyor, without doubling, as specified in 3.6.1. The capability of the machine to handle the specified load (see 1.2) within the time range specified in 3.5 and also the turnover requirement specified in 3.6.2 shall be verified.

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* 4.4.3 Electromagnetic interference test.- When electromagnetic compatibility is required, the preproduction sample or initial unit of production shall be tested by the supplier in accordance with test method CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or his private testing organization. The supplier shall furnish the contracting officer for approval the Interference Control Plan, the EMI/EMC Test Plan, and the Electromagnetic Compatibility Test Report required in MIL-STD-461. Upon approval of the test report by the contracting officer and provided all other requirements of the specification are met, the preproduction sample or initial unit of production as applicable, shall be used as a model for all production units.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B and C.- Each dough proofing machine shall be preserved, packaged, packed and marked for each level of protection in accordance with applicable requirements of MIL-B-2316 (see 6.2).

6. NOTES

6.1 Intended use.- The dough intermediate proofing machines covered by this specification are used in Garrison Bakeries to proof and convey loaf-sized rounds of various types of bread doughs from the dough rounding machine, MIL-D-2352, to the dough molding machine, MIL-D-2477.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, style and size required (see 1.2).
(c) First article.- When a preproduction sample is not required (see 3.1).
(d) Height of vertical leg, if other than specified (see 3.6.2).
(e) Electrical characteristics if other than specified (see 3.6.7).
(f) Manual contents and distribution of additional manuals (see 3.9).
(g) When electromagnetic interference control is required for Navy and Air Force use (see 3.10).
(h) Selection of applicable levels of preservation, packaging and packing (see 5.1).
(i) When the functional test (see 4.4.2 and 6.3) is waived.
(j) When supervision of installation is required (see 6.4).

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or slip clutch device *

6.3 Waiver of functional test.- The functional test described in 4.4.2 (see 6.2) may be waived provided that the supplier furnishes copies of certified test data covering previous tests of an identical model to the contracting officer. The test data must indicate compliance with the specified test conditions and must have been obtained as a result of tests conducted within the past 4 years on an identical type and rated dough proofing machine(s) furnished on a Government contract, and in addition to test data, a certified statement will be required.

6.4 Supervision.- Supervision of installation should be required of the supplier of dough proofing machine(s) shipped to any point in the Continental U.S., except those designated for storage or warehouse stock (see 6.2). When supervision of installation is required, the supplier shall furnish the services of a competent superintendent who shall be familiar with the construction, erection, operation, maintenance and repair of the equipment and the required utility connections. Material required for complete installations other than that specified herein and all installation labor will be furnished by the Government. All utilities will be extended and connected to the equipment by the Government.

6.5 Marginal notations.- The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Review activities:

Army - EL, MD
Navy - MC, SA, YD
Air Force - 84, 85

User activities:

Army - CE, ME

Preparing activity:

Army - GL

Project No. 7320-0272

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Dough Intermediate Proofing Machines - MIL-D-2351C

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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